

The Volume-Uniform Poincaré Wall: a Machine-Checked Obstruction for the Flat Block-Poincaré Route to Combes–Thomas Coercivity in Lattice Yang–Mills

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Abstract

Inside a Lean 4 formalization programme for four-dimensional $SU(N_c)$ lattice Yang–Mills, we report a machine-checked *negative* result and the machine-checked infrastructure that makes it meaningful. The positive substrate: (i) a fixed-volume Combes–Thomas chain for self-adjoint coercive finite-range lattice operators, instantiated on the flat gauge-fixed covariance of the physical shell, with coercivity constant $c = \min(1, a)/C_P$ fed by a proved fixed-volume flat Hodge/block-Poincaré inequality; and (ii) the concrete adjoint model of $SU(n) \rightarrow \mathfrak{su}(n)$ with the trace inner product, $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$, and the isometric transport to Euclidean coordinates — so that no statement below is discharged by a trivial model. The headline theorem is the *wall*: under the block normalization actually used by the formalized chain, every flat Hodge/block-Poincaré constant obeys $L^d/L^2 \leq C_P$ on the fine torus of side LN' , hence the volume-uniform Poincaré gate is *provably false* for $d \geq 3$, $N_c \geq 2$, and no positive coercivity constant survives all volumes through this route. The route consumed by the fixed-volume endpoint is therefore closed, by theorem rather than by impression. We further formalize the constant-sector quotient interface (the wall witness provably does not transfer to the fluctuation space; the quotient gate is neither proved nor refuted) and the exact witness family for its one-sided falsifier. Everything stated here is checked by Lean 4 against a pinned Mathlib, with zero `sorry`, zero project axioms, and a committed axiom-oracle transcript; a theorem–artifact map and reproduction instructions are included. No claim of progress on the Clay problem is made: the distance to the mass-gap problem is unchanged and effectively 0%.

1 Introduction

Constructive approaches to the four-dimensional Yang–Mills mass gap in the Balaban tradition [1, 2] route spectral information through renormalization-group covariances whose kernels must decay *uniformly in the volume*. A standard engine for such decay is the Combes–Thomas argument [3]: a self-adjoint coercive finite-range operator has an inverse with exponentially decaying kernel, at a rate governed by the coercivity constant. Whoever supplies the coercivity therefore controls the physics; and in a gauge-fixed lattice formalization the natural supplier is a Poincaré-type inequality tying the field norm to a Hodge term plus a block-averaging constraint.

This paper reports what happened when that plan was executed *formally*, in Lean 4 [6] against Mathlib [7], inside a long-running formalization programme whose earlier machine-

checked artifacts include a volume-uniform Wilson-loop area law via a formalized cluster expansion [8], rooted-tree majorants for polymer expansions with holes [9], a mechanized Bessel-ratio calculus for lattice gauge expansions [10, 11, 12], an exact two-dimensional SU(2) theory [13], and the audited programme map [15].

The outcome has three layers, and we state them with the labels they deserve.

Positive (fixed volume). A complete Combes–Thomas chain for coercive finite-range operators on finite metric graphs, instantiated at the flat gauge-fixed covariance of the physical shell: for every fixed volume there exist a positive rate θ and amplitude $2/c$ with $|K^{-1}(x, y)| \leq (2/c) e^{-\theta d(x, y)}$, where $c = \min(1, a)/C_P$ and C_P is a *proved* fixed-volume flat Hodge/block-Poincaré constant (Section 3).

Positive (non-vacuity). The abstract adjoint-model interface consumed by the chain is inhabited by the *true* matricial adjoint model of SU(n): $\mathfrak{su}(n)$ as the real inner-product space of traceless skew-Hermitian matrices under the trace form, its real dimension $n^2 - 1$, and the isometric transport to $\mathbb{R}^{n^2-1}$ carrying $\text{Ad}(g)X = gXg^\dagger$ (Section 4). No theorem below is satisfied merely by the trivial model.

Negative (the wall; the headline). Under the block normalization actually used by the formalized chain — the unscaled line-integral block map, which sends constants to L times constants — *every* flat Hodge/block-Poincaré constant obeys

$$\frac{L^d}{L^2} \leq C_P$$

on the fine torus of side $N = LN'$. Consequently (Section 5): the volume-uniform Poincaré gate is *false* for $d \geq 3$, $N_c \geq 2$; the Combes–Thomas coercivity fed by this route obeys $c \leq \min(1, a) L^2/L^d \rightarrow 0$; and no single $c_0 > 0$ is dominated at every volume by the route’s coercivity. The per-volume constants *do* exist, so the obstruction kills a real route, not an empty one.

Two further layers are *result-neutral* and are reported as such. The *constant-sector quotient interface* (Section 6) restricts the Poincaré question to the orthogonal complement of the constant sector; the wall witness provably does not transfer there, and the volume-uniform quotient gate is *open* — neither proved nor refuted. The *falsifier witness family* (Section 7) is an exact square-wave fluctuation mode intended for a one-sided test of that gate: it can refute the gate if its Rayleigh quotient degenerates; a bounded quotient on it would only eliminate one candidate. We are explicit about this asymmetry because the formal record must not overclaim in either direction.

Honest scope

This paper proves no mass gap, no volume-uniform Combes–Thomas bound, and nothing about the continuum limit; the distance to the Clay problem [5] is unchanged by these results and is effectively 0%. The wall is a theorem about *one specific route under one specific normalization* — precisely the route the formalized fixed-volume chain consumes. It does not exclude a volume-uniform Combes–Thomas estimate obtained after quotienting the harmonic sector, rescaling the block map (the wall must then be re-audited: it may merely move), or extracting coercivity from the interacting Wilson Hessian directly. The value of the negative result is navigational: it converts a would-be silent dead end into a machine-checked boundary, before any downstream construction is built on it.

2 The formal setting

All objects below are Lean definitions in the repository accompanying this paper (Section 9); we give their mathematical content and their Lean names.

Cochains. Fix $d, N, N_c \in \mathbb{N}$, all positive. Sites form the discrete torus $\Lambda_N = (\mathbb{Z}/N)^d$ (`FinBox d N`); positive bonds are pairs $b = (x, k)$ of a site and a direction (`PhysicalBond d N`). A *physical one-cochain* is a map A from positive bonds to the internal coordinate space $\mathbb{R}^{N_c^2-1}$ (`SUNLieCoord Nc`), with the ℓ^2 inner product in both the bond and internal indices (`PhysicalGaugeOneCochain d N Nc`).

The block map. For $N = LN'$, the flat block constraint Q : fine cochains (side LN') $\rightarrow$ coarse cochains (side N') assigns to a coarse bond the *unscaled* line integral (sum) of the L fine bond values along the corresponding block line (`flatBlockConstraintQCLM`, built on `linAvg`). The normalization fact that drives everything below is exact (`flatBlockConstraintQCLM_constant`):

$$Q(A_v) = L \cdot A_v^{\text{coarse}}, \quad (1)$$

where A_v denotes the direction-wise constant cochain $A_v(x, k) = v_k$ for a fixed family $v: \{1, \dots, d\} \rightarrow \mathbb{R}^{N_c^2-1}$ (`constantPhysicalGaugeOneCochain`).

The flat Hodge operator. At the trivial gauge background, $K_0 = D_1^\dagger D_1 + D_0 D_0^\dagger$ (`flatGaugeHodgeK0CLM`), so that $\langle A, K_0 A \rangle = \|\text{curl } A\|^2 + \|\text{div } A\|^2$ with the standard lattice curl on plaquettes and backward divergence on sites (`flatGaugeHodgeK0_inner_right`). Constants are flat-harmonic: $K_0 A_v = 0$ (`flatGaugeHodgeK0CLM_constantPhysicalGaugeOneCochain`).

The Poincaré predicate. For an adjoint model ρ (Section 4) and $C_P \in \mathbb{R}$, `FlatGaugeHodgePoincare d L N' Nc ρ C_P` is the statement

$$0 < C_P \quad \wedge \quad \forall A: \|A\|^2 \leq C_P (\langle A, K_0 A \rangle + \|QA\|^2), \quad (2)$$

over fine cochains at side LN' . The repository proves, at every fixed volume, that such constants exist (`exists_flatGaugeHodgePoincare`); this non-vacuity is what makes the negative results of Section 5 statements about a live route.

3 The fixed-volume Combes–Thomas endpoint

The Combes–Thomas chain is formalized for abstract self-adjoint coercive finite-range operators on finite metric graphs: the tilt algebra and tilted entry bounds, the block Schur bound, coercivity survival under tilting at the budget $M(e^{\theta R} - 1)N_R \leq c/2$, the tilted inverse with $\|K_\theta^{-1}\| \leq 2/c$, and the kernel extraction at a root (`PhysicalCoerciveCombesThomas*.lean`). Instantiated at the flat gauge-fixed covariance of the physical shell (stencils and locality proved, not assumed), the endpoint reads:

Theorem 3.1 (fixed-volume Combes–Thomas; `flatGaugeFixedCovariance_CT_fixedVolume`, `exists_flatGaugeFixedCovariance_CT_fixedVolume`). *Fix the volume parameters (d, L, N') ,*

a mass $a > 0$, and a flat Hodge/block-Poincaré constant C_P as in (2). Let $c = \min(1, a)/C_P$. Then there exists $\theta > 0$ meeting the explicit tilt budget such that the covariance kernel obeys

$$|K^{-1}(x, y)| \leq \frac{2}{c} e^{-\theta d(x, y)}.$$

Fixed volume means fixed volume: C_P , hence c and θ , may depend on (d, L, N', a) . The question the rest of this paper answers is whether *this* c can be made volume-uniform. It cannot.

4 The true adjoint model of $SU(n)$

The chain consumes an abstract interface `SUNAdjointModel n`: a family of isometries $\text{Ad}(g)$ of $\mathbb{R}^{n^2-1}$ indexed by $g \in SU(n)$ with $\text{Ad}(1) = \text{id}$, $\text{Ad}(gh) = \text{Ad}(g)\text{Ad}(h)$, and inner-product invariance. A trivial inhabitant ($\text{Ad}(g) = \text{id}$) exists and suffices for the flat lane, where only $\text{Ad}(1)$ is consumed; leaving matters there would invite a vacuity objection the moment a non-trivial background appears. The repository therefore constructs the genuine object:

Theorem 4.1 (the matricial adjoint model; `matrixSUNAdjointModel`). *Let $\mathfrak{su}(n)$ be the real vector space of traceless skew-Hermitian $n \times n$ complex matrices with the trace form $\langle X, Y \rangle = \text{Retr}(X^\dagger Y)$ (`SUNAdjointMatrixSubstrate`, `SUNAdjointInnerSpace`). Then $\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$ (`finrank_suLie`), the reindexed orthonormal-basis representation gives a linear isometric equivalence $\mathfrak{su}(n) \simeq \mathbb{R}^{n^2-1}$ (`suLieCoordIso`), and conjugating the adjoint action $X \mapsto gXg^\dagger$ by this isometry yields a `SUNAdjointModel n` whose action laws and inner-product invariance are proved, not hypothesized.*

The dimension count is itself formalized (Hermitian/skew decomposition, multiplication-by- i equivalence, ambient real dimension $2n^2$, and rank-nullity through the imaginary-trace functional with explicit surjectivity witness $i \cdot \mathbf{1}$). The coordinate choice is the non-canonical standard orthonormal basis; any other choice conjugates the model by a fixed orthogonal map, and nothing downstream depends on it.

5 The wall

Everything in this section is proved in `PhysicalGaugeFlatPoincare.lean` (the fixed-volume audit) and `PhysicalPoincareWall.lean` (the volume-uniform obstruction).

5.1 The constant sector forces the constant

Lemma 5.1 (exact sector normalization; `flatBlockConstraint_controls_constantSector`). *For every v ,*

$$\|A_v\|^2 = \frac{L^d}{L^2} \|QA_v\|^2.$$

Proof. $\|A_v\|^2 = N^d \sum_k \|v_k\|^2$ with $N = LN'$ (`norm_sq_constantPhysicalGaugeOneCochain`), while by (1), $\|QA_v\|^2 = L^2 N'^d \sum_k \|v_k\|^2$ (`flatBlockConstraintQCLM_constant_norm_sq`); divide. $\square$

Theorem 5.2 (the wall, fixed volume; `flatPoincare_constantSector_lower_bound`). *Let $N_c \geq 2$ and let C_P satisfy (2). Then*

$$\frac{L^d}{L^2} \leq C_P.$$

Proof. For $N_c \geq 2$ the internal space $\mathbb{R}^{N_c^2-1}$ is nontrivial, so there is v with $\sum_k \|v_k\|^2 > 0$ (`exists_pos_constantSector`; for $N_c = 1$, $\mathfrak{su}(1) = 0$ and the sector is empty — the hypothesis is stated, not hidden). Apply (2) to A_v : the Hodge term vanishes ($K_0 A_v = 0$), so $\|A_v\|^2 \leq C_P \|Q A_v\|^2$, and Lemma 5.1 gives the claim after cancelling $\|Q A_v\|^2 > 0$. $\square$

Corollary 5.3 (linear form; `flatPoincare_linear_lower_bound`). *If moreover $d \geq 3$, then $L \leq C_P$. (For $d = 4$ the forced growth is L^2 ; the linear form is all the divergence the obstruction needs. $d = 2$ is the scale-invariant exemption $L^d/L^2 = 1$ and is deliberately not claimed.)*

5.2 The route’s coercivity dies with the volume

Theorem 5.4 (coercivity upper bound; `flatPoincare_coercivity_upper_bound`). *For $a > 0$, $N_c \geq 2$, and any C_P as in (2),*

$$\frac{\min(1, a)}{C_P} \leq \min(1, a) \frac{L^2}{L^d}.$$

Theorem 5.5 (no volume-uniform coercivity through flat Poincaré; `no_volumeUniform_coercivity_via_flatP`). *Let $d \geq 3$, $N_c \geq 2$, $a > 0$. Suppose $c_0 \in \mathbb{R}$ is dominated at every volume: for every $L \in \mathbb{N}$ there exists $C_P(L)$ satisfying (2) at side $(L + 1)N'$ with $c_0 \leq \min(1, a)/C_P(L)$. Then $c_0 \leq 0$.*

Proof. If $c_0 > 0$, choose (Archimedes) n with $n > \min(1, a)/c_0$. By Corollary 5.3, $C_P(n) \geq n + 1 > \min(1, a)/c_0$, whence $\min(1, a)/C_P(n) < c_0$, contradicting domination at volume $n + 1$. $\square$

Theorem 5.6 (the gate is false; `volumeUniformFlatPoincareGate_false`). *For $d \geq 3$, $N_c \geq 2$, there is no single $C_P > 0$ satisfying (2) at every volume $L + 1$: the proposition `VolumeUniformFlatPoincareGate` — with the constant quantified before the volume — is provably false.*

Remark 5.7 (non-vacuity of the killed route; `perVolume_flatPoincare_family`). *The hypothesis family of Theorem 5.5 is inhabited: per-volume constants exist at every volume. The obstruction closes a route that is genuinely open at each fixed volume.*

Remark 5.8 (quantifier discipline). *The gate places its constant before the volume quantifier. The reversed order — for every volume there exists a constant — is exactly the (true) per-volume statement, and conflating the two is the standard way such obstructions get lost in prose. The formalization makes the distinction type-level.*

6 The constant-sector quotient interface

The wall witness lives in one sector. The natural continuation restricts the Poincaré question to its orthogonal complement — the physical fluctuation space. The repository formalizes the *interface* (`PhysicalPoincareSectorQuotient.lean`); we use the word “interface” deliberately: no quotient type or packaged orthogonal-complement submodule is constructed, and the predicate is a restriction to fluctuation cochains.

Definition 6.1 (fluctuation cochains; `IsFluctuationCochain`). A is a fluctuation cochain if $\langle A_v, A \rangle = 0$ for every constant generator A_v — equivalently, A is orthogonal to the span of the constant sector. (The generator-wise form is chosen for formal robustness.)

Proposition 6.2 (interface facts). (i) *The constant sector is flat-harmonic (`constantSectorLin_harmonic`) — which is why only the block term sees it, with the normalization of Lemma 5.1.* (ii) *The full predicate (2) implies its restriction to fluctuation cochains at the same constant (`quotientFlatPoincare_of_flatPoincare`); the restricted statement is strictly weaker, so Theorem 5.6 does not refute the quotient gate.* (iii) *Restricted constants exist at every fixed volume (`exists_quotientFlatPoincare`).* (iv) (Non-transfer.) *A nonzero constant cochain is not a fluctuation cochain (`constant_not_fluctuation`): the wall witness is excluded from the fluctuation space by construction.*

Remark 6.3 (the open gate, stated as open). *The volume-uniform quotient gate (`VolumeUniformQuotientPoincareGate`) is neither proved nor refuted in the formalization, and this paper makes no claim in either direction. Non-transfer removes the $W-1$ witness, not the possibility of a wall: slowly varying near-harmonic modes are the natural candidate counter-witness, which motivates the next section.*

7 The falsifier witness family

A *one-sided falsifier* for the quotient gate needs an explicit fluctuation family on which the quadratic form of (2) can be evaluated exactly. The repository provides the family and its two exact identities (`PhysicalPoincareLowModeFalsifier.lean`); the form evaluation itself (Hodge term, block term, Rayleigh quotient, endpoint decision) is registered future work, and we state the discipline that governs it.

Definition 7.1 (the lowest square-wave mode; `squareModeCochain`). On even side lengths $N = 2M$ ($M \geq 1$), with oscillation direction j , bond direction i , and internal vector w :

$$A(x, k) = \begin{cases} \varepsilon(x_j) w, & k = i, \\ 0, & k \neq i, \end{cases} \quad \varepsilon(m) = \begin{cases} +1, & m < M, \\ -1, & m \geq M, \end{cases}$$

(`squareSign`). The square profile is chosen over the cosine so that every identity below is an exact finite ± 1 computation; odd side lengths are out of scope by declared parameterization (a falsifier needs only *some* volume sequence).

Proposition 7.2 (exact identities). (i) $\sum_m \varepsilon(m) = 0$ (`sum_squareSign`), hence A is a fluctuation cochain (`squareModeCochain_isFluctuation`) — exact orthogonality, no approximation. (ii) $\|A\|^2 = (2M)^d \|w\|^2$ exactly (`norm_sq_squareModeCochain`). (iii) For $w \neq 0$ the mode equals no constant cochain (`squareModeCochain_ne_constant`), and the fluctuation space is nontrivial for $N_c \geq 2$ (`exists_nonzero_fluctuation`).

Remark 7.3 (one-sided discipline). *The registered continuation evaluates $R = (\langle A, K_0 A \rangle + \|QA\|^2) / \|A\|^2$ on this family, with no normalization that silently cancels L -dependence. Two outcomes are admissible: a volume sequence with $R \rightarrow 0$, which formally refutes the quotient gate (a second wall); or a uniform positive lower bound on this mode only, which eliminates one candidate and leaves the gate open. Only a spectral-minimality or all-modes argument could convert a favorable computation into a proof of the gate, and no such argument is claimed.*

8 Theorem–artifact map

Statement	Lean name	Module (YangMills/RG/)
CT endpoint, fixed volume	flatGaugeFixedCovariance_CT_fixedVolume	PhysicalShellCombesThomasEnc
CT positive-rate witness	exists_flatGaugeFixedCovariance_CT_fixedVolume	PhysicalShellCombesThomasEnc
$\dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1$	finrank_suLie	SUNAdjointDimension
Isometric transport	suLieCoordIso	SUNAdjointModelInstance
True adjoint model	matrixSUNAdjointModel	SUNAdjointModelInstance
Sector witness ($N_c \geq 2$)	exists_pos_constantSector	PhysicalPoincareWall
Wall, fixed volume	flatPoincare_constantSector_lower_bound	PhysicalPoincareWall
Wall, linear form ($d \geq 3$)	flatPoincare_linear_lower_bound	PhysicalPoincareWall
Coercivity upper bound	flatPoincare_coercivity_upper_bound	PhysicalPoincareWall
No volume-uniform c_0	no_volumeUniform_coercivity_via_flatPoincare	PhysicalPoincareWall
Gate false ($d \geq 3, N_c \geq 2$)	volumeUniformFlatPoincareGate_false	PhysicalPoincareWall
Per-volume non-vacuity	perVolume_flatPoincare_family	PhysicalPoincareWall
Fluctuation predicate	IsFluctuationCochain	PhysicalPoincareSectorQuotie
Full $\Rightarrow$ quotient	quotientFlatPoincare_of_flatPoincare	PhysicalPoincareSectorQuotie
Non-transfer of the witness	constant_not_fluctuation	PhysicalPoincareSectorQuotie
Quotient gate (open)	VolumeUniformQuotientPoincareGate	PhysicalPoincareSectorQuotie
Square profile, zero sum	sum_squareSign	PhysicalPoincareLowModeFalsi
Mode is fluctuation	squareModeCochain_isFluctuation	PhysicalPoincareLowModeFalsi
Mode norm, exact	norm_sq_squareModeCochain	PhysicalPoincareLowModeFalsi

Every listed name is exercised by the committed axiom-oracle script; on the frozen tree each prints exactly `[propext, Classical.choice, Quot.sound]`.

9 Reproducibility

Repository. <https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME> (branch `main`). The Lean tree described here is that of source checkpoint `12ca1a87`, sealed by commit `64969b35`; this manuscript is added in a strictly editorial later commit that changes no Lean file.

Toolchain. Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit `0764272048...` (lakefile and manifest agree). Build: `lake build YangMillsCore` — expected Build completed successfully (8410 jobs).

Axiom oracle. `lake env lean oracle_check.lean` executes 2194 `#print` axioms invocations; the committed transcript `docs/oracle-transcripts/ORACLE-20260714-12ca1a87.txt` records the full raw output with provenance header: 2174 results on a nonempty subset of `[propext, Classical.choice, Quot.sound]` (finely 2108 + 49 + 17) and 20 axiom-free results; zero `sorryAx`, zero nonstandard axiom sets. SHA-256 hashes (git blob form, LF): script `4376FA7E...`, raw output `3FAF5D32...`.

Verification ledger. The append-only ledger `docs/VERIFICATION-LEDGER.md` documents every checkpoint of the results here (Addenda 467–503), including the measurement-discipline rules (measure the pushed tree; blob-hash artifacts; positive-assertion counter sweeps) and the corrections history.

Continuation from this submission

The registered next steps, in order: evaluation of the Hodge term on the witness family (two jump slices); evaluation of the block term and the Rayleigh quotient; the endpoint decision for the quotient gate (second wall or candidate-eliminated); and, independently of the outcome, the two other registered routes (rescaled block map with re-audited sector normalization; interacting-Hessian coercivity). Each will be reported with the same discipline — exact identities, declared parameterizations, one-sided claims only where only one side is proved.

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